

Input Instruction	写一篇关于美国政治新闻的博客文章，标题用双角括号标示，即«标题»，并且确保句子数量少于5句（不包括5句）。句子应当较长，以使你的回答总字数达到或超过250字。 Write a blog post about American political news with the title indicated by double angle brackets, i.e. «Title», and ensure that the number of sentences is less than 5. The sentences should be relatively long to make the total number of words in your response reach 250.
Model Response	«美国政治新闻：分裂与变革的交响曲» 在2022年的美国，政治新闻如同一部紧张的交响曲，旋律中交织着分裂与变革的强烈节奏。民主党和共和党的分歧日益加剧，疫苗接种的鸿沟、经济复苏的不平等以及社会正义的呼唤都在琴弦上弹出激昂的和弦。川普总统的退场并未平息争议，而是开启了拜登政府的新篇章，他的“重建美国”计划在国会中遭遇了重大挑战，立法进程中的僵局成为新闻焦点。与此同时，地方选举中的基层力量崛起，显示选民对传统两党制的不满正在发酵。这场政治交响乐，每个音符都预示着未来走向的不确定性，但无论如何，美国政治的戏剧性永远不会缺少观众。 «American Political News: Symphony of Division and Transformation» In 2022, American political news is like a tense symphony, with a melody interwoven with the strong rhythms of division and change. The divide between the Democratic and Republican parties is intensifying, and the chasms in vaccine distribution, unequal economic recovery, and calls for social justice all play passionate chords on the strings. The departure of President Trump has not quelled the controversy but has opened a new chapter for the Biden administration. His "Build Back Better" plan has faced significant challenges in Congress, with the deadlock in the legislative process becoming a focus of the news. Meanwhile, grassroots forces are rising in local elections, showing that voters' dissatisfaction with the traditional two-party system is fermenting. Every note in this political symphony foretells the uncertainty of the future, but no matter what, the drama of American politics will never lack an audience.
Scoring Question	输出内容是否是关于美国政治新闻的博客文章？（满足用户需求） Is the model response a blog post about American political news? (Helpfulness)
Evaluator Output	分析：根据模型回复的内容，可以看到它确实是一篇关于美国政治新闻的博客文章。开头使用了双角括号标示的标题《美国政治新闻：分裂与变革的交响曲》，接着以较长的句子描述了美国政治的现状，提及了与政治相关的关键词汇，如民主党、共和党、川普总统、拜登政府、立法进程、地方选举等。因此，模型回复确实是一篇关于美国政治新闻的博客文章。 答案：是 Analysis: Based on the content of the [Model Response], it can be seen that it is indeed a blog post about American political news. The response starts with a title enclosed in double angle brackets, "American Political News: Symphony of Division and Transformation," followed by longer sentences describing the current state of American politics, mentioning keywords related to politics such as Democrats, Republicans, President Trump, the Biden administration, legislative processes, local elections, etc. Therefore, the [Model Response] is indeed a blog post about American political news. Answer: Yes
Evaluation Result	1 (Yes)

Table 12: An example of automatic evaluation with evaluator LLM and their English translation.

F Detailed Information about Human Annotation

We recruited 12 college students for data annotation of COMPLEXBENCH. As for annotators training, all the annotators are required to complete a training tutorial that includes 50 samples. We provide feedback to help them calibrate the annotation criteria. In addition, during the annotation process, we conduct spot checks on annotated data. We also retrain the annotators based on the issues found in their annotations and require them to recheck such issues in their annotated data. The annotator training process for dataset creation and human evaluation is similar. And the total labor cost is approximately 2800\$. We provide the guidelines for each annotation task as follows.

F.1 Guidelines for Data Annotation

This section corresponds to Section 4.1 (**Data Annotation and Validation**), where the guidelines for data annotation are shown in Table 13, and Table 14 presents the English translation version. In this annotation task, annotators will be provided with reference instructions, the requirements of the minimum number of constraint dimensions in each constraint type, and the minimum number of composition types. Annotators are instructed to construct new complex instructions based on the reference instructions while annotating all the constraint dimensions and composition types within the newly constructed instructions. Then, they are also required to annotate scoring questions for the newly constructed instructions, and the task type of the newly constructed instructions.

F.2 Guidelines for Selection Branch Expansion

This section corresponds to Section 4.1 (**Selection Branch Expansion**), where the guidelines for selection branch expansion are shown in Table 15, and Table 16 presents the English translation version. In this annotation task, instructions with *Selection* that are annotated in the above task will be provided to the annotators. Annotators are instructed to modify the selection conditions of the original instructions and construct several new instructions to cover all the different selection branches apart from the original instructions. For each new instruction, all the information required by the above annotation task needs to be annotated.

F.3 Guidelines for Overall Preference Annotation

This section corresponds to Section 5.1 (**Agreement Evaluation**), where the guidelines for overall preference annotation and their English translation are shown in Table 17. Given an instruction and two model responses (denoted as A and B), the human annotators are instructed to compare the quality and choose from 3 options, namely A better than B, tie, and B better than A.

F.4 Guidelines for Scoring Questions Verification

This section corresponds to Section 5.1 (**Agreement Evaluation**), where the guidelines for scoring questions verification and their English translation are shown in Table 18. Given an instruction and a corresponding model response, as well as a scoring question for the instruction, human annotators are instructed to judge whether the requirements of the scoring question are satisfied by the model response.

F.5 Guidelines for Instruction Decomposition

This section corresponds to Section 5.2.3 (**Decomposition of instructions with composition types**), where the guidelines for instruction decomposition and their English translation are shown in Table 19. Given an instruction containing composition types, human annotators are instructed to decompose the instruction based on composition types (e.g., *Chain* into sequential tasks, *Selection* into selection and execution branches, while *And* remains intact) and split the scoring questions of original instructions into corresponding decomposed instructions.

G An Example of Data Construction

We provide a specific example (translated into English) about data annotation in Table 20 to facilitate a better understanding of the process. The objectives of the first two steps in data construction, i.e.,

Reference Instruction Collection and **Task Allocation** (as mentioned in Section 4.1) are to obtain **Reference Instruction** and **Task Requirements** fields of the table, respectively. Each field of the table corresponds to Table 13.

下面将给你提供一条参考指令，四个约束类型各自的约束维度数量要求，以及组合方式数量要求，请你依次完成如下标注任务。

1. 基于参考指令构造出一条新的复杂指令，要求该指令中包含的各约束类型的约束维度数量大于或等于要求数量，该指令中包含的各组合方式的数量大于或等于要求数量。你也可以不参考参考指令从头开始编写。
2. 标注新构造的复杂指令的任务类型，在十个类型中选择最接近的一类。
3. 标注新构造的复杂指令中包含的所有约束维度，组合方式。
4. 标注新构造的复杂指令的得分问题。请针对该指令中的每一个约束维度、组合方式，分别设计一个可以用“是/否”回答的得分问题判定其是否得到满足。
5. 标注得分问题之间的依赖关系。我们规定，对于链式组合方式，后续任务的所有得分问题依赖于判断前序任务是否完成的得分问题；对于选择组合方式，判断正确分支执行情况的所有得分问题依赖于判断正确分支是否被选择的得分问题。在标注每个得分问题时，同时需要标注其依赖于哪些得分问题。

在构造复杂指令时，必须遵循如下三个总体原则：

1. 合理性：指令必须没有歧义，有相对明确正确答案。
2. 约束有效性：指令中包含的每一个约束维度，均应该对输出产生实质影响。
3. 难度：指令必须是足够困难的，原则上应该比参考指令更为复杂。

[参考指令]

{reference_instruction}

[任务要求]

词级约束最少数量：{number_of_lexical_constraints}

格式约束最少数量：{number_of_format_constraints}

语义约束最少数量：{number_of_semantic_constraints}

整体约束最少数量：{number_of_utility_constraints}

并列组合最少数量：{number_of_And}

链式组合最少数量：{number_of_Chain}

选择组合最少数量：{number_of_Selection}

请你根据参考指令构造出新的复杂指令：{newly_constructed_instruction}

请选择构造指令的任务分类，从以下十项中选择一项：{task_type_of_newly_constructed_instruction}

A. 基本能力 B. 中文理解 C. 综合问答 D. 实用文本写作 E. 创意写作 F. 专业文本写作 G. 个性化写作 H. 逻辑推理 I. 角色扮演 J. 专业能力

请选择构造指令中包含的所有词级约束，多选，一个选项可以选择多次：{lexical_constraints_in_newly_constructed_instruction}

A. 输入词匹配 B. 输出关键词

请选择构造指令中包含的所有格式约束，多选，一个选项可以选择多次：{format_constraints_in_newly_constructed_instruction}

A. Json格式 B. Markdown格式 C. 分点格式 D. 标点格式 E. 输出长度 F. 开头格式 G. 结尾格式 H. 基于模板格式

请选择构造指令中包含的所有语义约束，多选，一个选项可以选择多次：{semantic_constraints_in_newly_constructed_instruction}

A. 语言风格 B. 角色属性 C. 话题 D. 情感

请选择构造指令中包含的所有整体约束，多选，一个选项可以选择多次：{utility_constraints_in_newly_constructed_instruction}

A. 目标语言 B. 支持性 C. 连贯性 D. 事实正确性 E. 满足用户需求

请选择构造指令中包含的所有组合方式，多选，一个选项可以选择多次：{composition_types_in_newly_constructed_instruction}

A. 并列 B. 链式 C. 选择

请标注构造指令的所有评分问题，同时标注出其评测的约束维度/组合方式，每个得分问题参考如下例子编写：

1. 模型生成的文章语言是否为英文？（目标语言）

{scoring_questions_for_newly_constructed_instruction}

请标注评分问题之间的依赖关系，每个得分问题参考如下例子编写（没有依赖关系则不必编写）：

4. 模型回复字数是否在300字左右？（输出长度，依赖于1）

{dependencies_of_scoring_questions}

Table 13: Guidelines for data annotation. The blue part is the information provided to the annotators, and the red part is content that requires the annotators to make annotations.

H English Translation of COMPLEXBENCH

Since the data we collected from real-world application scenarios is primarily in Chinese, we initially use our data construction pipeline to construct Chinese data. Subsequently, we use GLM-4[41] to translate the constructed data into English and manually correct any errors in the translation to produce the English version of COMPLEXBENCH.